

Comparative Study of Mentor-Mentee Matching Algorithms

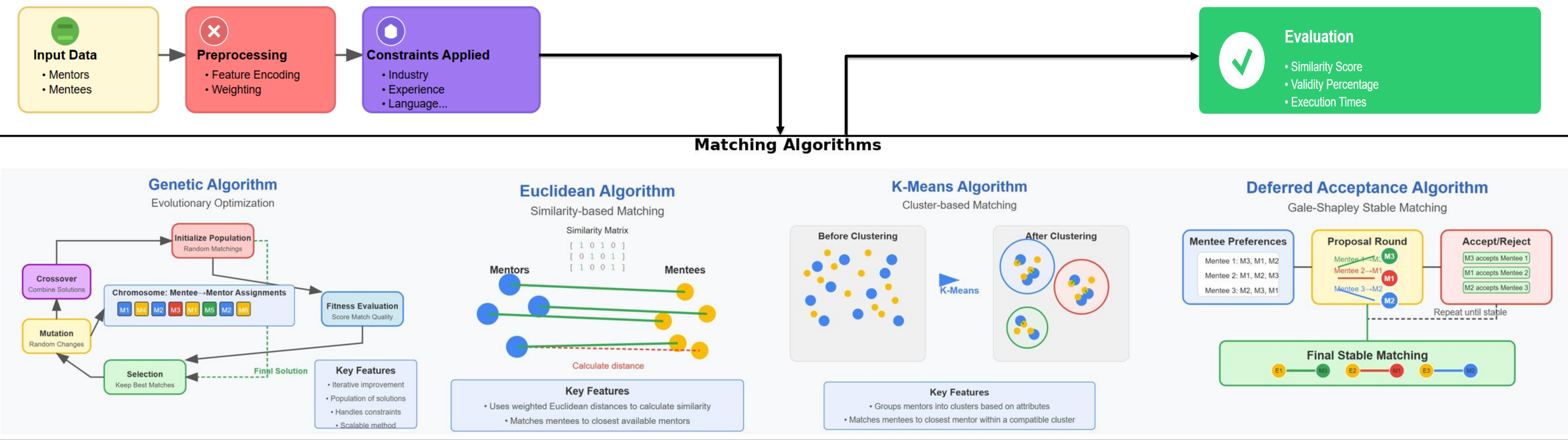
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Introduction

Effective mentor-mentee matching is critical for professional development but remains a challenging optimisation problem. Current manual or simple rule-based methods often fail to account for the complex, multidimensional factors that contribute to successful mentoring relationships. This research builds on previous work by Shah et al. (2024) to comparatively analyse four distinct algorithmic approaches: Euclidean distance-based matching, K-means clustering, Genetic Algorithm (GA), and Deferred Acceptance (DA).

Research Questions

1. What is the impact of two-sided matching compared to one-sided?
2. How does adding constraints affect validity, similarity and execution time?
3. Can simple algorithms compete with complex ones for constrained matching?
4. How early do algorithms break down under increasing constraints?



Experimental Setup

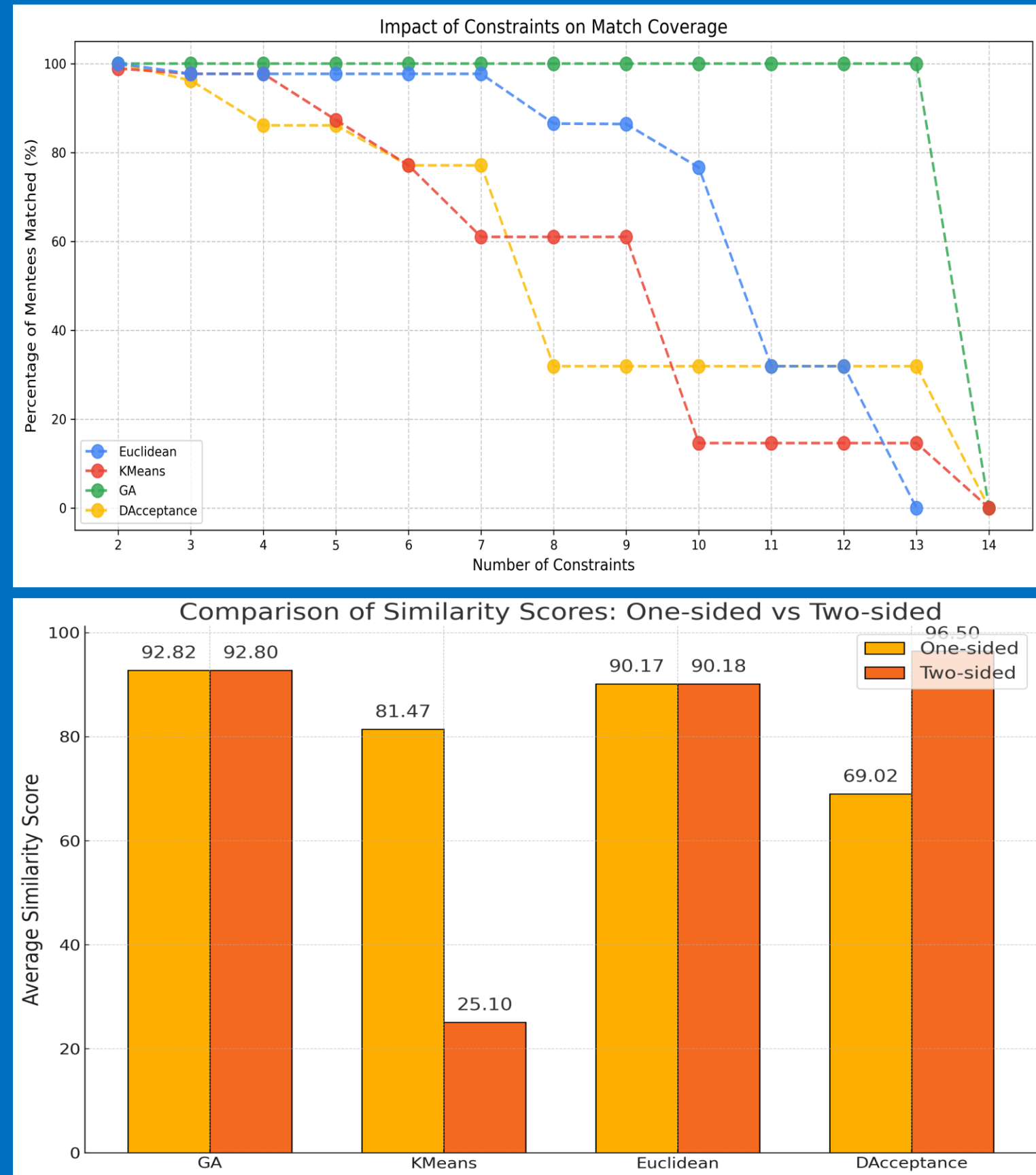
- **Data:** 1000 mentees and 500 mentors randomly sampled from 8000
- **Matching Features:** 13 per person
- **Algorithms Runs:** 100 randomized each
- **Constraints Applied:** 18 randomized
- **Clusters:** 10 groups
- **Mutation Rate:** 0.01
- **Crossover Rate:** 0.9
- **Generations:** 20
- **Population Size:** 100
- **Tournament Size:** 5

Conclusions

- **Euclidean** achieved the best balance of speed and match quality, even as constraints increased.
- **Deferred Acceptance (DA)** maintained 100% validity under all constraints but was slower.
- **Genetic Algorithm (GA)** had the highest similarity scores but became slower and less stable with more constraints.
- **K-Means** was fast early on but broke down quickly with increasing constraints.
- **Two-sided matching** reduced similarity and increased runtime compared to one-sided, especially under complex constraints.
- **Simple algorithms** like Euclidean can outperform complex ones when constraint load is manageable.
- **Future Work:** Combine Euclidean's speed with fairness from DA, and explore adaptive constraint handling for better robustness.

Key Results

Algorithm	Constraints	Validity %	Similarity Score	Total Matches	Valid Matches	Execution Time (ms)
Euclidean (Onesided)						
	2.00	89.52	90.17	1000.00	595.23	38.01
	3.00	100.00	86.10	948.00	948.00	87.78
	4.00	100.00	86.10	948.00	948.00	90.88
	5.00	100.00	85.50	945.79	945.79	137.15
	6.00	100.00	85.50	945.79	945.79	140.07
	7.00	100.00	83.98	771.00	771.00	218.99
	8.00	100.00	83.98	771.00	771.00	218.19
	9.00	100.00	83.98	771.00	771.00	214.59
	10.00	100.00	83.98	771.00	771.00	202.47
	11.00	100.00	83.98	771.00	771.00	201.31
	15.00	100.00	77.91	148.00	148.00	524.35
	20.00	0.00	0.00	0.00	0.00	585.47
Kmeans (Onesided)						
	2.00	53.17	25.09	1000.00	531.57	155.10
	3.00	50.65	24.85	688.33	348.66	240.73
	4.00	50.65	24.85	688.33	348.66	243.47
	5.00	45.05	23.57	682.19	307.50	292.40
	6.00	61.48	23.87	563.87	345.57	427.86
	8.00	27.55	23.40	54.79	15.08	988.73
	10.00	21.50	24.28	30.43	6.69	1009.03
	15.00	0.00	0.00	0.00	0.00	1023.98
GA (Onesided)						
	2.00	99.99	92.82	538.91	538.72	4500.20
	3.00	97.28	92.48	494.74	481.18	3893.91
	5.00	96.18	91.99	473.34	455.28	3538.83
	10.00	94.27	91.55	428.03	401.82	2443.54
	15.00	93.66	91.40	408.99	381.09	2133.47
	16.00	93.73	91.38	404.45	379.08	1900.70
DA (Onesided)						
	2.00	92.80	96.50	1000.00	928.04	38.37
	3.00	92.80	96.50	1000.00	928.04	43.98
	4.00	96.21	96.27	977.00	940.00	58.16
	6.00	100.00	95.59	873.00	873.00	117.86
	10.00	100.00	92.18	771.00	771.00	198.09
	15.00	100.00	92.18	771.00	771.00	200.75
	16.00	100.00	84.97	610.00	610.00	318.80
	19.00	100.00	78.15	148.00	148.00	517.78
	20.00	0.00	0.00	0.00	0.00	581.13



References and Acknowledgements

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Acknowledgements

The author would like to sincerely thank: **Jing Hua Ye** and **Ruairí O'Reilly**